Summary Test Maths

This covers sections of Algebra (A1 to A5) from the edexcel specification.

Topics tested (Use Maths Genie to learn if needed):

1. Simplifying Algebra (Q1 - Q6)
2. Indices - (Q7 - Q13)
3. Substitution (Q14)
4. Rearranging Harder Formulae (Q15 - Q18)
5. Expanding Triple Brackets (Q19 - Q22)

Q1.

Simplify $3x + 4x - 2x$

Q2.

Simplify $8g + 6h - 3g + h$

Q3.

(a) Simplify $a \times b \times c$

(b) Simplify $5p - 2p$

(c) Simplify $\frac{6h}{3}$

Q4.

(a) Simplify $2a \times 3b$

(b) Simplify $2p \times 2p$

(c) Simplify $\frac{7x + 5x}{4}$

Q5.

(a) Simplify $6j \times 5k$

(b) Simplify $7a - 6b + 5a + 4b$

Q6.

(a) Simplify $a^2 + a^2 + a^2$

(b) Simplify $2rs - 5rs + 4rs$

(c) Simplify $4a + 2 - 7a + a - 6$

Q7.

(a) Simplify $x^8 \times x^3$

(b) Simplify $(5y)^3$

(c) Simplify $\dfrac{w^7}{w^4}$

Q8.

(a) Simplify $2m^2 \times 5n^6$

(b) Simplify $15p^3 \div 3p^4$

Q9.

$y^2 \times y^a = y^7$

(a) Find the value of a.

$(y^4)^b = y^{12}$

(b) Find the value of b.

Q10.

Work out the value of $\dfrac{2^9 \times 2^{-2}}{2^4}$

Q11.

(a) Simplify $9p^3 \times 2p^{-2}$

Q12.

(a) Simplify $5c^2d^3 \times 2d$

(b) Write 64×4^5 as a power of 4

Q13.

Simplify $\dfrac{10p^3q^5r}{4p^3q^6}$

Q14.

$$s = \frac{v^2 - u^2}{2a}$$

$v = 7$
$u = 5$
$a = 3$

Work out the value of s.

Q15.

Make a the subject of the formula $v = u + at$

Q16.

Make v the subject of the formula $T = \frac{1}{2}mv^2$

Q17.

Make x the subject of the formula $a = \frac{x + 4}{x + 2}$

Q18.

Make x the subject of the formula $\frac{a}{b} = \frac{2x}{x + 5}$

Q19.

Expand and Simplify $(x + 2)(x + 4)(x + 1)$

Q20.

Expand and Simplify $(x + 3)(x - 1)^2$

Q21.

Expand and Simplify $(2x - 3)(x - 2)(3x - 1)$

Q22.

Show that $(5x - 4)(3x + 1)(2x - 7) = 30x^3 - 119x^2 + 41x + 28$

for all values of x.

Answer :

A1.

Simplify $3x + 4x - 2x$

$7x - 2x$

$5x$

A2.

Simplify $8g + 6h - 3g + h$

$5g + 7h$

A3.

(a) Simplify $a \times b \times c$

abc

(1)

(b) Simplify $5p - 2p$

$3p$

(1)

(c) Simplify $\frac{6h}{3}$

$2h$

(1)

A4.

(a) Simplify $2a \times 3b$

$6ab$

(1)

(b) Simplify $2p \times 2p$

$4p^2$

(1)

(c) Simplify $\frac{7x + 5x}{4}$

$\frac{12x}{4}$

$3x$

(1)

A5.

(a) Simplify $6j \times 5k$

$30jk$

(1)

(b) Simplify $7a - 6b + 5a + 4b$

$12a - 2b$

(2)

A6.

(a) Simplify $a^2 + a^2 + a^2$

$3a^2$

(1)

(b) Simplify $2rs - 5rs + 4rs$

$-3rs + 4rs$

rs

(1)

(c) Simplify $4a + 2 - 7a + a - 6$

$-2a - 4$

(2)

A7.

(a) Simplify $x^8 \times x^3$

x^{11}

(1)

(b) Simplify $(5y)^3$

$5y \times 5y \times 5y$

$125y^3$

(1)

(c) Simplify $\frac{w^7}{w^4}$

w^3

(1)

A8.

(a) Simplify $2m^2 \times 5n^6$

$10m^2n^6$

(1)

(b) Simplify $15p^3 \div 3p^4$

$5p^{-1}$

(2)

A9.

$y^2 \times y^a = y^7$

(a) Find the value of a.

5

(1)

$(y^4)^b = y^{12}$

(b) Find the value of b.

3

(1)

A10.

Work out the value of $\dfrac{2^9 \times 2^{-2}}{2^4}$

$$\frac{2^7}{2^4} = 2^3 = 8$$

$\cancel{2^3}$ 8

A11.

(a) Simplify $9p^3 \times 2p^{-2}$

18p

A12.

(a) Simplify $5c^2d^3 \times 2d$

$10c^2d^4$

(1)

(b) Write 64×4^5 as a power of 4

$$4^3 \times 4^5 = 4^8$$

4^8

A13.

Simplify $\dfrac{10p^3q^5r}{4p^3q^6}$

$$\frac{\cancel{10}^{5}\cancel{p^3}\cancel{q^5}r}{\cancel{4}_{2}\cancel{p^3}q^{\cancel{6}}} \qquad \frac{5r}{2q}$$

$\dfrac{5r}{2q}$

A14.

$$s = \frac{v^2 - u^2}{2a}$$

$v = 7$
$u = 5$
$a = 3$

Work out the value of s.

$$s = \frac{(7)^2 - (5)^2}{2(3)}$$

$$= \frac{49 - 25}{6}$$

$$= \frac{24}{6}$$

$$= 4$$

$s =$ 4

A15.

Make a the subject of the formula $v = u + at$

$$-u \quad -u$$

$$\frac{v - u}{t} = \frac{at}{t}$$

$$a = \frac{v - u}{t}$$

A16.

Make v the subject of the formula $T = \frac{1}{2}mv^2$

$$2T = mv^2$$

$$\frac{2T}{m} = v^2$$

$$\sqrt{\frac{2T}{m}} = v$$

A17.

Make x the subject of the formula $a = \frac{x+4}{x+2}$

$$a(x+2) = x+4$$

$$ax + 2a = x + 4$$

$$-x \qquad -x$$

$$ax - x + 2a = 4$$

$$-2a \qquad -2a$$

$$ax - x = 4 - 2a$$

$$x(a-1) = 4 - 2a$$

$$x = \frac{4-2a}{a-1}$$

A18.

Make x the subject of the formula $\frac{a}{b} = \frac{2x}{x+5}$

$$a(x+5) = 2x(b)$$

$$ax + 5a = 2bx$$

$$5a = 2bx - ax$$

$$5a = x(2b - a)$$

$$x = \frac{5a}{2b - a}$$

A19.

Expand and Simplify $(x+2)(x+4)(x+1)$

$$(x+2)(x^2 + x + 4x + 4)$$

$$(x+2)(x^2 + 5x + 4)$$

$$x^3 + 5x^2 + 4x + 2x^2 + 10x + 8$$

$$x^3 + 7x^2 + 14x + 8$$

A20.

Expand and Simplify $(x+3)(x-1)^2$

$$(x+3)(x-1)(x-1)$$

$$(x+3)(x^2 - x - x + 1)$$

$$(x+3)(x^2 - 2x + 1)$$

$$x^3 - 2x^2 + x + 3x^2 - 6x + 3$$

$$x^3 + x^2 - 5x + 3$$

A21.

Expand and Simplify $(2x-3)(x-2)(3x-1)$

$$(2x-3)(3x^2-x-6x+2)$$

$$(2x-3)(3x^2-7x+2)$$

$$6x^3-14x^2+4x-9x^2+21x-6$$

$$\underline{6x^3-23x^2+25x-6}$$

A22.

Show that $(5x-4)(3x+1)(2x-7) = 30x^3 - 119x^2 + 41x + 28$

for all values of x.

$$(5x-4)(6x^2-21x+2x-7)$$

$$(5x-4)(6x^2-19x-7)$$

$$30x^3-95x^2-35x-24x^2+76x+28$$

$$\underline{30x^3-119x^2+41x+28}$$

Summary Test Biology

This covers sections of 4.1 - Cell Biology (All of 4.1.1 - Cell Structure) from the AQA specification.

Topics tested:

4.1.1.1 Eukaryotes and prokaryotes
4.1.1.2 Animal and plant cells
4.1.1.3 Cell specialisation
4.1.1.4 Cell differentiation
4.1.1.5 Microscopy

Time: 50 minutes
Total Marks: 40

Section A: Core Knowledge (10 marks)

1. State two differences between eukaryotic and prokaryotic cells. (2)

_____

_____

2. Describe the function of: (3)

a) Ribosomes

b) Mitochondria

c) Cell membrane

3. Explain why bacterial cells are described as prokaryotic. (2)

_____

4. Give one example of a specialised animal cell and describe how it is adapted to its function. (3)

_____

Section B: Application and Microscopy (15 marks)

5. A plant cell is observed under a microscope. It contains chloroplasts and a large vacuole.

Explain how these structures help the plant cell function effectively. (4)

_____

6. Calculate the actual size of a cell if the image size is 2.4 mm and magnification is ×600.

Show full working. (3)

7. Explain why electron microscopes provide greater detail than light microscopes. (4)

_____

8. Describe how cell differentiation differs in animals and plants. (4)

_____

Section C: Higher Challenge (15 marks)

9. Compare a bacterial cell and a plant cell in terms of structure and genetic material. (6)

---

10. A student observes cells dividing rapidly in the root tip of a plant.

Explain why cells in this region show high levels of differentiation later in development. (4)

---

11. A microscope has a ×10 eyepiece and a ×40 objective lens.

a) Calculate total magnification. (1)

b) Explain how increasing magnification can affect resolution. (4)

Mark Scheme

1. Any two: nucleus present in eukaryotes; DNA free in cytoplasm in prokaryotes; prokaryotes smaller; plasmids present. (2)

2. Ribosomes – protein synthesis (1); Mitochondria – aerobic respiration/energy release (1);

Cell membrane – controls movement of substances in and out (1)

3. Genetic material not enclosed in a nucleus. (2)

4. Example e.g. sperm cell – tail for movement; nerve cell – long axon; red blood cell – no nucleus, biconcave. (3)

5. Chloroplasts – photosynthesis (2); Vacuole – maintains turgor/support (2)

6. Convert mm to µm if needed; 2.4 mm = 2400 µm; 2400 ÷ 600 = 4 µm. (3)

7. Electrons have shorter wavelength; higher resolution; greater magnification; more detail visible. (4)

8. Animals differentiate early and are limited; plants differentiate throughout life at meristems. (4)

9. 6-mark comparison: nucleus vs none; plasmids; cell wall composition; chloroplasts absent in bacteria; size differences; cytoplasm/ribosomes. (6)

10. Meristem cells divide then specialise; gene expression changes; adaptation for specific functions. (4)

11. a) 400 (1); b) Magnification does not equal resolution; higher magnification without resolution gives blurred image; resolution = ability to distinguish detail. (4)